Lab 07: VLAN Segmentation

Segmenting a LAN with Virtual LANs on a Cisco Switch

Lab #7 **Topic:** VLANs & Switching **Tool:** Cisco Packet Tracer **Cert:** CompTIA Network+

1. Lab Objectives

- Create multiple VLANs on a Cisco Catalyst switch.
- Assign switch ports to specific VLANs (access ports).
- Verify that devices in different VLANs cannot communicate without a router.
- Configure a trunk port to carry multiple VLANs to a router.
- Relate to Network+ objective: 2.3 – Explain the purposes of VLANs.

2. Network Topology

One Cisco 2960 switch with three VLANs. Each VLAN represents a logical segment. PCs in the same VLAN can communicate; PCs in different VLANs are isolated at Layer 2.

VLAN ID	VLAN Name	Ports	IP Subnet	PCs
10	SALES	Fa0/1–Fa0/5	192.168.10.0/24	PC1, PC2
20	TECH	Fa0/6–Fa0/10	192.168.20.0/24	PC3, PC4
30	MGMT	Fa0/11–Fa0/15	192.168.30.0/24	PC5

3. Step-by-Step Configuration

3.1 Create VLANs on the Switch

```
Switch> enable
Switch# configure terminal
Switch(config)# vlan 10
Switch(config-vlan)# name SALES
Switch(config-vlan)# exit

Switch(config)# vlan 20
Switch(config-vlan)# name TECH
Switch(config-vlan)# exit

Switch(config)# vlan 30
```

```
Switch(config-vlan)# name MGMT
Switch(config-vlan)# exit
```

3.2 Assign Ports to VLANs (Access Mode)

```
Switch(config)# interface FastEthernet0/1
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
Switch(config-if)# exit
```

```
Switch(config)# interface FastEthernet0/6
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 20
Switch(config-if)# exit
```

```
Switch(config)# interface FastEthernet0/11
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 30
Switch(config-if)# exit
Switch(config)# exit
Switch# write memory
```

4. Verification

```
Switch# show vlan brief            ! Lists all VLANs and their ports
Switch# show interfaces trunk      ! Shows trunk ports (if configured)
```

```
PC1> ping 192.168.10.12  ! Same VLAN → SUCCESS
PC1> ping 192.168.20.10  ! Diff VLAN → FAIL (no routing)
```

5. Key Takeaways

- VLANs create logical broadcast domain separation on a single physical switch.
- Access ports carry traffic for ONE VLAN only; trunk ports carry multiple VLANs using 802.1Q tagging.
- Without a Layer 3 device (router or Layer 3 switch), VLANs cannot communicate with each other.
- VLANs improve security (isolate departments), reduce broadcast traffic, and simplify management.
- The **show vlan brief** command is the fastest way to verify VLAN port assignments.